

– For online publication only –

Exposure to the COVID-19 Stock Market Crash and its Effect on Household Expectations

Survey Instructions

Tobin Hanspal* Annika Weber[†] Johannes Wohlfart[‡]

1. Introduction

Welcome

Thank you very much for participating in this study!

This study is conducted by researchers from the University of Copenhagen, Goethe University Frankfurt and the WU Vienna University of Economics and Business.

Participants will be asked to answer a few questions about their opinions as well as a set of demographic questions. Participation in the study typically takes 15 minutes and is strictly anonymous.

In case you experience any technical problems please do not restart the survey but contact us at johannes.wohlfart@econ.ku.dk. Feel free to use the same email address in case you have any question or concern about the survey content.



*Tobin Hanspal, Department of Finance, Accounting and Statistics, WU Vienna University of Economics and Business, e-mail: tobin.hanspal@wu.ac.at

[†]Annika Weber, Department of Finance, Goethe University Frankfurt, e-mail: annika.weber@hof.uni-frankfurt.de

[‡]Johannes Wohlfart, Department of Economics and CEBI, University of Copenhagen, CESifo, e-mail: johannes.wohlfart@econ.ku.dk

Please note that each person is only allowed to **participate in the survey once**. The survey cannot be interrupted and restarted. Thus, if you encounter a technical problem, please do not restart the survey. Also, **complete the survey all in one go** and **avoid any disturbances**.

Moreover, we would like to ask you to **not use any other functionalities of your computer** during the survey such as listening to music or browsing the web. Finally, we ask you to respond to the survey **alone**. All these requirements are essential for the quality of the research project.

Informed consent form

On the next page, you will find an informed consent form.



Consent form:

I have read the information provided on the previous page.

I have had the opportunity to ask questions about the study.

I understand that I may withdraw from the study at any time.

I understand how to raise a concern or make a complaint.

I understand that I can only participate in this survey once.

I understand that close attention to the survey is required for my responses to count.

If you are 18 years of age or older, agree with the statements above, and freely consent to participate in the study, please choose "I agree" to begin the survey.

I agree.



I disagree.



The next question is about the following problem. In questionnaires like ours, sometimes there are participants who do not carefully read the questions and just quickly click through the survey. This means that there are a lot of random answers which compromise the results of research studies.

To show that you read our questions carefully, please choose both “Very strongly interested” and “Not at all interested” as your answer in the next question.

Given the above, how interested are you in sports?

☐ Very strongly interested

☐ Very interested

☐ A little bit interested

☐ Almost not interested

☐ Not at all interested



2. Basic Demographics

In this survey we will ask you at various times about things related to your household, such as the total income of your household.

By household we mean all family members living with you in your main residence, but excluding roommates and renters.



Which of these describes you more accurately?

☐ Male

☐ Female

What is your age?

☐ 18-24 years

☐ 25-34 years

☐ 35-44 years

☐ 45-54 years

☐ 55-64 years

☐ 65 years and older

In which state do you currently reside?

What is the highest level of education you have completed?

☐ 12th grade or less

☐ Graduated high school or equivalent

☐ Some college, no degree

☐ Associate degree

☐ Bachelor's degree

☐ Post-graduate degree

What was the **total gross income** of all members of your household (including you), from all sources before taxes and deductions, in 2019?

- ☐ \$14,999 and less
- ☐ \$15,000 - \$24,999
- ☐ \$25,000 - \$49,999
- ☐ \$50,000 - \$74,999
- ☐ \$75,000 - \$99,999
- ☐ \$100,000 - \$149,999
- ☐ \$150,000 - \$199,999
- ☐ \$200,000 and more
- ☐ Prefer not to say

Which of these describes your labor market situation **at the end of January 2020, i.e. before the start of the current crisis**, most accurately?

- ☐ Employed full-time
- ☐ Employed part-time
- ☐ Self-employed
- ☐ Unemployed
- ☐ Retired
- ☐ Student
- ☐ Homemaker
- ☐ Other:



3. Household Assets and Liabilities as of January 2019

Now we would like to ask you several questions about your household's wealth.

Please note that these data are **completely anonymized**, and we are not able to personally link you to your answers in the survey. In case you are nevertheless unwilling to respond to the following questions, please indicate "prefer not to say".



Please think of all your household's **retirement accounts**, including 401(K)s, IRAs, or Keogh accounts.

About how much were your household's investments in these accounts worth ***at the end of January 2020, i.e. before the start of the current crisis?***

If your household did not have any retirement accounts, please select \$0.

☐ \$0
☐ \$1 - \$999
☐ \$1,000 - \$4,999
☐ \$5,000 - \$9,999
☐ \$10,000 - \$24,999
☐ \$25,000 - \$49,999
☐ \$50,000 - \$74,999
☐ \$75,000 - \$99,999
☐ \$100,000- \$124,999
☐ \$125,000- \$149,999
☐ \$150,000- \$174,999
☐ \$175,000- \$199,999
☐ \$200,000 - \$249,999
☐ \$250,000 - \$499,000
☐ \$500,000 and more
☐ Prefer not to say.



Please think of all your household's **other investable financial assets** (outside of retirement accounts).

What was the total value of these assets **at the end of January 2020, i.e. before the start of the current crisis?**

Please include the value of all your household's bank accounts, brokerage accounts, etc. Please do NOT include the value of any real estate property or any private business you own. Please do also **NOT include** the value of your **retirement accounts you already reported above**.

If your household did not have any other investable financial assets, please select \$0.

☐ \$0
☐ \$1 - \$999
☐ \$1,000 - \$4,999
☐ \$5,000 - \$9,999
☐ \$10,000 - \$24,999
☐ \$25,000 - \$49,999
☐ \$50,000 - \$74,999
☐ \$75,000 - \$99,999
☐ \$100,000- \$124,999
☐ \$125,000- \$149,999
☐ \$150,000- \$174,999
☐ \$175,000- \$199,999
☐ \$200,000 - \$249,999
☐ \$250,000 - \$499,000
☐ \$500,000 and more
☐ Prefer not to say.



About what percent of the investments in your household's **retirement accounts** was in stocks or stock mutual funds *at the end of January 2020*?

percent.

And about what percent of your household's **other investments in financial assets** (outside of retirement accounts) was in stocks or stock mutual funds *at the end of January 2020*?

percent.



4. Losses to Financial Wealth and Household Net Income

Overall, would you say that the combined value of all your household's **retirement accounts**, including 401(K)s, IRAs, or Keogh accounts, has decreased due to the recent drop in stock prices?

☐ Yes

☐ No



By about what percent did the combined value of your household's retirement accounts **decrease** as a result of the recent stock market decline since January, before the start of the current crisis, as of today?

percent.



Note: Participants are asked by what percentage the value of their retirement accounts did *increase* if they indicated "No" in the question on whether the value of their retirement assets decreased and "Yes" in a subsequent question on whether the value of their retirement assets increased.

Overall, would you say that the combined value of all your household's **other investable financial assets** (outside of retirement accounts, such as bank accounts, brokerage accounts, etc.) has decreased due to the recent drop in stock prices?

☐ Yes

☐ No



By about what percent did the value of your household's other investable financial assets (outside of retirement accounts) **decrease** as a result of the recent stock market decline since January, before the start of the current crisis, as of today?

percent.



Note: Participants are asked by what percentage the value of their other investable financial assets did *increase* if they indicated "No" in the question on whether their financial assets decreased and "Yes" in a subsequent question on whether their financial assets increased.

We now would like to know how your personal employment situation and household's income are affected by the current crisis.

Did you lose your job **since the beginning of 2020** as a result of the current crisis?

☐ Yes

☐ No



Was the **total net income** of all members of your household (including you) **since the beginning of 2020** higher or lower than you had expected at the beginning of the year, i.e. before the start of the current crisis?

(Total net income includes all sources of income in the household, after taxes and deductions and including transfers.)

☐ Higher

☐ Lower

☐ About the same



By about what percentage was your household's net income **lower** than what you had expected? Please give your best guess.

percent.



Note: Participants are asked by what percentage their household's net income was *higher* if they indicated "Higher" in the previous question.

5. Prior Beliefs about the Duration of Stock Market Recoveries

5.1 "Black Monday" Arm

Now we would like you to think about the drop in stock prices during **the stock market crash of 1987**, also known as the **"Black Monday"**.

What do you think, **how many years** did it take until US stock prices had recovered to **their pre-crisis level** (starting from the beginning of the drop in stock prices in August 1987)?



5.2 "Dot-Com" Arm

Now we would like you to think about the drop in stock prices after **the burst of the Dot-com Bubble in 2000**.

What do you think, **how many years** did it take until US stock prices had recovered to **their pre-crisis level** (starting from the beginning of the drop in stock prices in August 2000)?



5.3 "Financial Crisis" Arm

Now we would like you to think about the drop in stock prices during the **Financial Crisis** that started in **2007**.

What do you think, **how many years** did it take until US stock prices had recovered to **their pre-crisis level** (starting from the beginning of the drop in stock prices in October 2007)?



6. Information Treatment

6.1 "Black Monday" Arm

On the next page we will provide you with information on the drop in stock prices **after the stock market crash of 1987**, also known as the **"Black Monday"**.

Please **review the information carefully**.

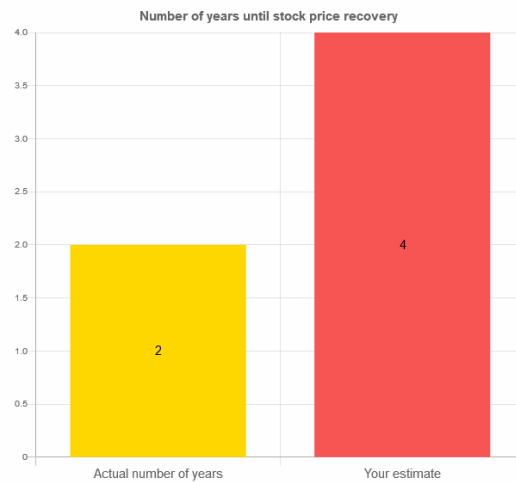
The information will be shown to you only once, and you will **not be able to come back** to it.



You estimated that stock prices recovered to their pre-crisis level **4** years after the beginning of their drop in August 1987.

We now provide you with information on the actual development of stock prices after the "Black Monday" in 1987.

Starting from the beginning of the drop in stock prices in August 1987, it took **about 2 years** until stock prices recovered to their pre-crisis level in July 1989.



Source: The number is based on the time that the stock market index S&P500 needed to recover to its pre-crisis level.

Note: You will be able to go to the next slide after you have spent a few seconds on this slide.



6.2 "Dot-Com" Arm

On the next page we will provide you with information on the drop in stock prices after **the burst of the dot-com bubble in 2000**.

Please **review the information carefully**.

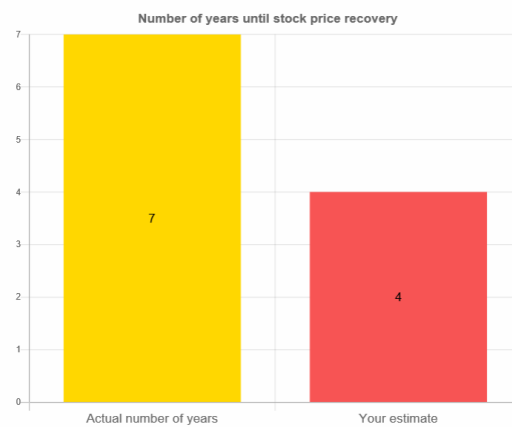
The information will be shown to you only once, and you will **not be able to come back to** it.



You estimated that stock prices recovered to their pre-crisis level **4** years after the beginning of their drop in August 2000.

We now provide you with information on the actual development of stock prices after the burst of the dot-com bubble in 2000.

Starting from the beginning of the drop in stock prices in August 2000, it took **about 7; years** until stock prices recovered to their pre-crisis level in June 2007.



Source: The number is based on the time that the stock market index S&P500 needed to recover to its pre-crisis level.

Note: You will be able to go to the next slide after you have spent a few seconds on this slide.



6.3 "Financial Crisis" Arm

On the next page we will provide you with information on the drop in stock prices during the Financial Crisis that started in 2007.

Please **review the information carefully**.

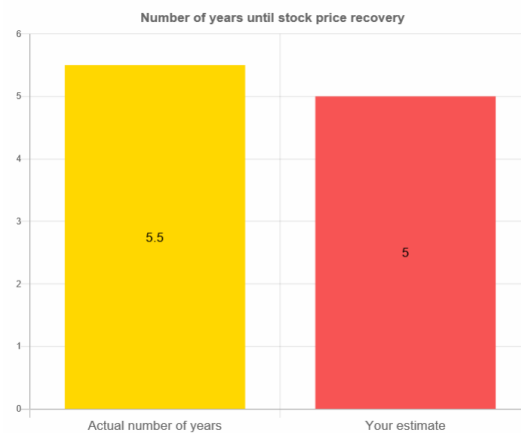
The information will be shown to you only once, and you will **not be able to come back** to it.



You estimated that stock prices recovered to their pre-crisis level **5** years after the beginning of their drop in October 2007.

We now provide you with information on the actual development of stock prices during the Financial Crisis that started in 2007.

Starting from the beginning of the drop in stock prices in October 2007, it took **about 5 ½ years** until stock prices recovered to their pre-crisis level in March 2013.



Source: The number is based on the time that the stock market index S&P500 needed to recover to its pre-crisis level.

Note: You will be able to go to the next slide after you have spent a few seconds on this slide.



7. Posterior Beliefs and Behavioral Measures

Now please think again about the **current** situation.

What do you think, in **which calendar year** will the US stock market have recovered to **the level of January 2020** (the level before the recent drop in stock prices)?



To what extent do you agree with the following statements?

"The outbreak of the coronavirus will keep US stock prices below their January 2020 levels for many years."

Strongly agree	Agree	Somewhat agree	Neither agree nor disagree	Somewhat disagree	Disagree	Strongly disagree
☐	☐	☐	☐	☐	☐	☐

"The outbreak of the coronavirus has set the level of the stock market back by many years."

Strongly agree	Agree	Somewhat agree	Neither agree nor disagree	Somewhat disagree	Disagree	Strongly disagree
☐	☐	☐	☐	☐	☐	☐

"The US stock market will have recovered by the end of the year 2020."

Strongly agree	Agree	Somewhat agree	Neither agree nor disagree	Somewhat disagree	Disagree	Strongly disagree
☐	☐	☐	☐	☐	☐	☐



What do you think, **in which calendar year** will your own household's **net wealth** have recovered to the level of January 2020?

(Your household's net wealth is the difference in value between all your household's assets and all your household's debts.)

Please think about the development of the stock market and the overall economy as well as your own household's situation and decisions regarding saving, spending, and work in the time to come.

☐ Year:

☐ My own household's net wealth will never recover to its pre-crisis level.



In this question we present you with eight possible scenarios for the **US stock market return** over the next 12 months.

(Note: This expected return is the change in value, in percent, that you expect to receive over the next 12 months from investing in a portfolio that holds all stocks listed on the US stock market. It includes both dividends and capital gains/losses.)

Please let us know what probability you attach to each scenario. The probabilities of the eight scenarios need to sum to 100 percent.

Probability (in percent) that the US stock market return will be ...

... less than -30 percent	0
... between -30 and -15 percent	0
... between -15 and -5 percent	0
... between -5 and 0 percent	0
... between 0 and 5 percent	0
... between 5 and 15 percent	0
... between 15 and 30 percent	0
... greater than 30 percent	0
Total	0



In general, do you think your household will be financially better or worse off **12 months from now** than you are these days?

☐ Much worse off
☐ Worse off
☐ Somewhat worse off
☐ About the same
☐ Somewhat better off
☐ Better off
☐ Much better off



In general, do you think your household will be financially better or worse off **12 months from now** than you are these days?

☐ Much worse off

☐ Worse off

☐ Somewhat worse off

☐ About the same

☐ Somewhat better off

☐ Better off

☐ Much better off



Now think about your **household's total spending**, including groceries, clothing, personal care, housing (such as rent, mortgage payments, utilities, maintenance, home improvements), medical expenses (including health insurance), transportation, recreation and entertainment, education, and any large items (such as home appliances, electronics, furniture, or car payments).

Compared to 2019, I expect my **total household spending** in the year **2020** to be...

☐ ... higher.

☐ ... lower.

☐ ... about the same.



By about what percent do you expect your total household spending to be **higher**?

Please give your best guess.

percent.



In 2020, what do you expect will happen to the **total net income** of all members of your household (including you), from all sources after taxes and deductions and including transfers?

Compared to 2019, I expect my **total net household income** in **2020** to be...

☐ ... higher.

☐ ... lower.

☐ ... about the same.



By about what percent do you expect your total net household income to be **lower**?

Please give your best guess.

percent.



What do you think, in **which calendar year** will your own household's **net income** have recovered to the level of 2019?

(Total net income includes all sources of income in the household, after taxes and deductions and including transfers.)

Please think about the development of the overall economy as well as your own household's situation and decisions, for instance regarding work, in the time to come.

☐ Year:

☐ My own household's net income will never recover to its pre-crisis level.



How are your expectations regarding your **retirement age** affected by the current crisis?

☐ I will certainly retire later than I had expected before the crisis.

☐ I may retire later than I had expected before the crisis.

☐ Not affected.

☐ I may retire earlier than I had expected before the crisis.

☐ I will certainly retire earlier than I had expected before the crisis.

How are your expectations regarding your **desired working hours** in the next years affected by the current crisis?

☐ I will certainly try to work more than I had expected before the crisis.

☐ I may try to work more than I had expected before the crisis.

☐ Not affected.

☐ I may try to work less than I had expected before the crisis.

☐ I will certainly try to work less than I had expected before the crisis.



How are your expectations regarding your household's **total debt** (including mortgage debt, credit card balances, etc.) that will be outstanding at the end of 2020 affected by the current crisis?

- ☐ My household's debt will be a lot higher than I had expected before the crisis.
- ☐ My household's debt will be somewhat higher than I had expected before the crisis.
- ☐ Not affected.
- ☐ My household's debt will be somewhat lower than I had expected before the crisis.
- ☐ My household's debt will be a lot lower than I had expected before the crisis.



The next question is about potential active adjustments to the share of stocks and stock mutual funds in total financial wealth (**both inside and outside of retirement accounts**) your household may have made **since the start of the current crisis**.

Which of the following describes your household's recent actions best?

My household has **actively, through purchases and sales of stocks or stock mutual funds**,...

- ☐ ... strongly increased the share of stocks/stock mutual funds in financial wealth.
- ☐ ... increased the share of stocks/stock mutual funds in financial wealth.
- ☐ ... somewhat increased the share of stocks/stock mutual funds in financial wealth.
- ☐ ... somewhat decreased the share of stocks/stock mutual funds in financial wealth.
- ☐ ... decreased the share of stocks/stock mutual funds in financial wealth.
- ☐ ... strongly decreased the share of stocks/stock mutual funds in financial wealth.
- ☐ My household has not made any active adjustments to the share of stocks/stock mutual funds in financial wealth.



Over the next weeks, do you expect your household to make active adjustments to the share of stocks and stock mutual funds in total financial wealth (**both inside and outside of retirement accounts**)?

I expect my household to actively, through purchases and sales of stocks or stock mutual funds,...

- ☐ ... strongly increase the share of stocks/stock mutual funds in financial wealth.
- ☐ ... increase the share of stocks/stock mutual funds in financial wealth.
- ☐ ... somewhat increase the share of stocks/stock mutual funds in financial wealth.
- ☐ ... somewhat decrease the share of stocks/stock mutual funds in financial wealth.
- ☐ ... decrease the share of stocks/stock mutual funds in financial wealth.
- ☐ ... strongly decrease the share of stocks/stock mutual funds in financial wealth.
- ☐ I do not expect my household to make any active adjustments to the share of stocks/stock mutual funds in financial wealth.



How difficult would it currently be for your household to **obtain a loan of \$1,000**, e.g. by taking out consumer credit or using a credit card or line of credit?

- ☐ Very difficult
- ☐ Somewhat difficult
- ☐ Neither easy nor difficult
- ☐ Somewhat easy
- ☐ Very easy



8. Additional Household Characteristics

Did you **lose wealth** that was invested in the stock market due to the **Financial Crisis** that began in **2007**?

- ☐ Did not lose any stock market wealth.
- ☐ Lost some stock market wealth.
- ☐ Lost substantial stock market wealth.

Did you **lose wealth** that was invested in the stock market due to the **burst of the Dot-com Bubble in 2000**?

- ☐ Did not lose any stock market wealth.
- ☐ Lost some stock market wealth.
- ☐ Lost substantial stock market wealth.

Did you **lose wealth** that was invested in the stock market due to the stock market crash of the **"Black Monday" in 1987**?

- ☐ Did not lose any stock market wealth.
- ☐ Lost some stock market wealth.
- ☐ Lost substantial stock market wealth.



Some people are very **knowledgeable about personal finances**, while others are less knowledgeable about personal finances.

On a scale from zero to ten, where zero is not at all knowledgeable about personal finance and ten is very knowledgeable about personal finance, what number would you be on the scale?

0	1	2	3	4	5	6	7	8	9	10
☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐

For how many years has your household been investing in stocks or stock mutual funds?



Some people are fully **prepared to take financial risks** when they save or make investments, while others try to avoid taking financial risks.

On a scale from zero to ten, where zero is not at all willing to take risks and ten is very willing to take risks, what number describes your **general willingness to take financial risks** best?

0	1	2	3	4	5	6	7	8	9	10
☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐



On the scale below, which of the following best describes how **financial decisions** are made in **your household**?

Someone else in my household makes all financial decisions		I share financial decisions equally with someone else in my household		I make all financial decisions myself
1	2	3	4	5
☐	☐	☐	☐	☐



In January 2020, were you the main contributor of income in your household?

- ☐ Yes
- ☐ No
- ☐ My spouse and I contributed equally.
- ☐ Other, please specify:



Please tell us how many people usually live in your current primary residence, including yourself, and excluding non-relatives like roommates or renters?

What is the zip code of your current main residence?

What is your marital status?

☐ Single

☐ Married

☐ Divorced

☐ Widowed

☐ Separated

Which category would best describe your political orientation?

☐ Democrat

☐ Republican

☐ Other

☐ Prefer not to say

What is your year of birth?



At the end we would like to ask you some final questions on your household's wealth and income in January 2020, before the start of the current crisis.

What was the total value of the **real estate** your household owned **at the end of January 2020, i.e. before the start of the current crisis**? That is, without taking any outstanding loans into account, about how much would you have earned, had it been sold at the end of January 2020?

If your household did not own any real estate, please select \$0.

- ☐ \$0
- ☐ \$1 - \$49,999
- ☐ \$50,000 - \$99,999
- ☐ \$100,000 - \$199,999
- ☐ \$200,000 - \$299,999
- ☐ \$300,000 - \$399,999
- ☐ \$400,000 - \$499,999
- ☐ \$500,000 - \$749,999
- ☐ \$750,000 - \$999,999
- ☐ \$1,000,000 - \$1,499,999
- ☐ \$1,500,000 and more
- ☐ Prefer not to say.



What was the total value of all **debt** your household had outstanding ***at the end of January 2020, i.e. before the start of the current crisis?***

Please include any mortgages, lines of credit, education loans, vehicle loans, other installment loans, and credit card balances outstanding.

If your household did not have any debt outstanding, please select \$0.

☐ \$0
☐ \$1 - \$999
☐ \$1,000 - \$4,999
☐ \$5,000 - \$9,999
☐ \$10,000 - \$24,999
☐ \$25,000 - \$49,999
☐ \$50,000 - \$74,999
☐ \$75,000 - \$99,999
☐ \$100,000 - \$199,999
☐ \$200,000 - \$299,999
☐ \$300,000 - \$499,999
☐ \$500,000 - \$749,999
☐ \$750,000 and more
☐ Prefer not to say.

What was the **total net income** of all members of your household (including you), from all sources after taxes and deductions and including transfers, in 2019?

- ☐ \$14,999 and less
- ☐ \$15,000 - \$24,999
- ☐ \$25,000 - \$49,999
- ☐ \$50,000 - \$74,999
- ☐ \$75,000 - \$99,999
- ☐ \$100,000 - \$149,999
- ☐ \$150,000 - \$199,999
- ☐ \$200,000 and more
- ☐ Prefer not to say



Did you look up anything on the internet while you were taking this survey, e.g. using a search engine such as Google?

- ☐ Yes
- ☐ No



What did you look up, and which website(s) or search engine(s) did you use?



What do you think was the **purpose of this survey**? 1-2 sentences should be enough.



Thank you very much for completing our survey!

If there are any remarks that you would like to make or clarifications that you would like to obtain, please do let us know by writing them into the field below:

